GIS-Based Land Suitability Analysis for Urban Development using Land Use Land Cover: Bengaluru South

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Live demo: iamabhinav-dev.github.io/gis-urban-suitability

Abstract

Where can new urban development reasonably go? We answer this question for a 582 km^2 area of south Bengaluru, India, entirely in Python, from open 10 m land-cover maps (2018–2023), a 30 m DEM and OpenStreetMap. Land is classified into built-up, forest and usable by K-Means clustering of 100 m cells (96.6 % agreement with the source classes). For any existing building, the usable cells whose surroundings (land cover in 250 and 500 m rings, distances to built-up land, roads and water, terrain) are most similar are found and explained. A building-similarity score and a random forest trained on recent change are combined into a suitability score, graded into classes and grouped into candidate sites. Every score is computed with data as of 2018 and tested against land that became built in both 2022 and 2023, under strict leakage rules and next to simple baselines. The combined score reaches an AUC of 0.797 (top-decile lift 3.45), ahead of the distance to built-up land (0.763) and a classic AHP overlay (0.636); its two best classes hold 65 % of the growth. Growth next to existing built-up land is predicted well, growth away from it not at all.

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1 Introduction

Indian cities are growing fast into the land around them. Bengaluru is a clear case: over four decades its built-up area grew by 584 %, while vegetation fell by 66 % and water bodies by 74 % (Ramachandra et al., 2012). Planners therefore keep facing the same question: *where can new urban development reasonably go?*

The classic GIS answer is a land suitability analysis. Factor maps such as slope, access to roads and current land cover are rated, weighted and overlaid into one suitability surface (Collins et al., 2001). Since the 1990s the weights usually come from multi-criteria decision analysis (MCDA), most often the Analytic Hierarchy Process (AHP; Saaty, 1980), in which experts compare criteria in pairs (Malczewski, 2004). The method is transparent and easy to explain, but it has two well-known weaknesses. The weights are subjective, and changing only the weighting method moves the boundaries of the suitable zones (Mosadeghi et al., 2015). And the resulting maps are rarely checked against what actually happened on the ground (Malczewski, 2004; Parry et al., 2018).

Freely available satellite land use / land cover (LULC) data now make a different approach possible. Annual 10 m LULC maps (Karra et al., 2021) show where land has actually been built on, year by year. A model can learn which surroundings preceded new development, and its predictions can be tested against growth that happened later.

This project builds such an analysis for **Bengaluru South**, entirely in Python. It answers three questions, set at the project meeting:

1. **Land classification.** Which land is built-up, which is forest, and which is usable (potential) land? The land is grouped by clustering LULC fractions and terrain on a 100 m grid.

2. **Building similarity.** Given an existing building at location A, which locations B on usable land have similar surroundings (terrain, access to roads and water, nearby land cover, built-up density)?

3. **Suitability.** How do data-driven scores (similarity to existing buildings and a random forest that learns from recent change) compare with a classic MCDA/AHP overlay when all of them are tested against real growth?

The test is strict: every score is computed with data as of 2018 only and compared with growth that persisted through 2022 and 2023. Rules against leakage of future information are enforced in code. The result is a three-class land map, a graded suitability map with candidate sites, and an honest comparison of the methods, including where they fail.

Section 2 reviews related work. Sections 3 and 4 describe the study area and the data. The methods and the validation design follow in Sections 5 and 6, the results in Section 7, and a discussion and conclusion close the report.

2 Literature review

Two families of methods rank land for urban development: expert MCDA and learned suitability. Table 1 summarises the papers reviewed; the full notes are in `docs/literature_review.md`.

2.1 Expert multi-criteria suitability

Land-use suitability analysis began with hand-drawn overlays and moved to GIS, while its core idea stayed the same: map the factors, rate them, and overlay them into a composite (Collins et al., 2001). Malczewski (2004) groups GIS methods into overlay mapping, MCDA (weighted linear combination, often with AHP weights) and soft-computing approaches. He notes that results depend heavily on subjective choices of criteria, standardisation and weights, and that suitability maps are rarely validated. The weighting method alone matters: on the same criteria, AHP

and Fuzzy-AHP ranked development options the same way but drew different zone boundaries (Mosadeghi et al., 2015). A typical Indian study, Parry et al. (2018), combines slope, altitude, LULC and the status of existing amenities with AHP weights at ward level for Srinagar and Jammu, without validation.

The criteria are similar across studies: terrain (slope, elevation), current land cover, and proximity to roads, existing built-up land and water. Our feature table has the same core, at 100 m cells instead of wards, and our MCDA baseline uses exactly these criteria.

2.2 LULC change in Bengaluru and Indian cities

Ramachandra et al. (2012) classified Landsat images of Bangalore from 1973 to 2010: built-up area grew 584 % while vegetation fell 66 % and water bodies 74 %. Growth has eaten into exactly the land our exclusion mask protects (lakes and forest). Across five other Indian mega cities, Bharath et al. (2018) find that industrial, infrastructural and socio-economic drivers influence growth more than biophysical factors. This matches our design, in which road access and nearby built-up land are the main context features and terrain is secondary.

2.3 Learning suitability from data

A second line of work learns a suitability surface from where development already is or recently happened: neural networks (Pijanowski et al., 2002; Liu et al., 2017), random forests (Ahmadlou et al., 2016; Li et al., 2024) and presence-only maximum entropy (Wang et al., 2021). They share a core of predictors with ours: distances to roads, rivers and existing urban land, terrain, and neighbourhood land use.

Validation is the weak point. Studies that test on a random split of a single date report very high accuracy, such as AUC 0.99 for urban suitability in Yulin (Li et al., 2024), where the cell's own land use is also an input. Studies that test against *later observed change* report more modest values, such as AUC 0.82 in Rasht (Ahmadlou et al., 2016). Random cross-validation over-states performance on spatially structured data (Roberts et al., 2017; Ploton et al., 2020); spatial blocks give more honest estimates (Valavi et al., 2019). Growth type also matters: growth away from existing urban land is much harder to predict than growth next to it (Wang et al., 2021), which can be measured with the Landscape Expansion Index (Liu et al., 2010). The total operating characteristic complements ROC for maps where change is a small share of the land (Pontius Jr. and Si, 2014).

Data quality shapes the design as well. Among the global 10 m LULC products, the Esri map has the highest overall accuracy, but it over-estimates scrub (Venter et al., 2022). OSM building completeness varies strongly between cities and over time (Herfort et al., 2023).

2.4 The gap this project addresses

Expert MCDA is transparent but subjective and almost never tested against real growth. Learned models remove the hand-set weights but are often validated optimistically. This project:

- tests *every* score, including the expert MCDA baseline, the same way: against persistent growth from 2018/19 to 2022/23, always next to simple baselines (random and distance to built-up);
- enforces leakage rules: no own-cell land cover in model inputs, OSM data as of the baseline date, a time-travel rule, and out-of-fold scores with spatial blocks;
- adds a per-building similarity query ("find land whose surroundings resemble building A"), which none of the reviewed papers offers.

Table 1: Papers reviewed. Validation: how the paper tests its result.

Paper	Area	Method	Validation
Malczewski (2004)	review	GIS-MCDA methods	rarely done (noted)
Collins et al. (2001)	USA (review)	history of suitability analysis	—
Mosadeghi et al. (2015)	Gold Coast	AHP vs Fuzzy-AHP overlay	two outputs compared
Parry et al. (2018)	Srinagar, Jammu	GIS-AHP, ward level	none reported
Ramachandra et al. (2012)	Bangalore	Landsat LULC change, 1973–2010	classification accuracy
Bharath et al. (2018)	5 Indian cities	CA-Markov with growth agents	not reported in the abstract
Li et al. (2024)	Yulin	random forest, 25 factors	random 70/30 split, AUC 0.99
Wang et al. (2021)	Luoyang	MaxEnt + cellular automaton	2009→2018, Kappa 0.78
Ahmadlou et al. (2016)	Rasht	random forest	later change, AUC 0.82, TOC
Liu et al. (2017)	China	neural network + CA (FLUS)	2000→2010 simulation
Pijanowski et al. (2002)	Michigan	neural network (LTM), 100 m	same period
Roberts et al. (2017), Ploton et al. (2020)	methods	blocked cross-validation	random vs blocked CV
Liu et al. (2010), Pontius Jr. and Si (2014)	methods	LEI growth types; TOC curve	—
Venter et al. (2022), Herfort et al. (2023)	global	LULC accuracy; OSM completeness	ground truth; ML estimate

3 Study area

The study area is **Bengaluru South**, a 582 km^2 rectangle from 77.52° E to 77.74° E and 12.72° N to 12.94° N (Figure 1). It covers the south-eastern growth corridor along Hosur Road (Electronic City, Bommasandra), the Kanakapura Road fringe in the south-west, and **Bannerghatta National Park**, of which 41.3 km^2 lies inside the area. All analysis is done in UTM zone 43N (EPSG:32643) on a grid of 58,218 cells of 100 m.

Why this area. Fourteen candidate areas of 400–620 km^2 around Indian cities were compared on their land-cover mix, their persistent growth since 2018 and their OpenStreetMap coverage (`docs/study_area.md`). Bengaluru South was chosen for four reasons:

- **strong, persistent growth:** 8.4 % of the land that was non-built in 2018 was built in both 2022 and 2023, which gives enough change to validate against;
- **a clear forest block** (Bannerghatta) next to cropland and scrub, which gives the three-class map a real forest class;
- **mostly gentle terrain:** elevations of 727–1022 m (median 898 m) and a median cell slope of 2.5°; under 1 % of the cells are steeper than 15°, mostly in the Bannerghatta hills;
- **good OpenStreetMap coverage in 2018:** about 151,000 building footprints were already mapped on 1 January 2018. Several other candidates (Coimbatore, Bhubaneswar) were mapped in bulk only later, so they could not supply buildings as of the baseline date, which the temporal validation needs.

Land cover. In 2018 the ESRI map classes 53.6 % of the area as built, 18.4 % as crops, 18.1 % as rangeland, 7.2 % as trees and 2.5 % as water. By 2023 built land had risen to 61.5 %, mostly at

the expense of cropland (12.2 %). The built-up core lies in the north, toward central Bengaluru; growth is concentrated in the south and east. Many small lakes (tanks) are scattered across the area, and some of them are overgrown with vegetation, which the exclusion mask has to handle.

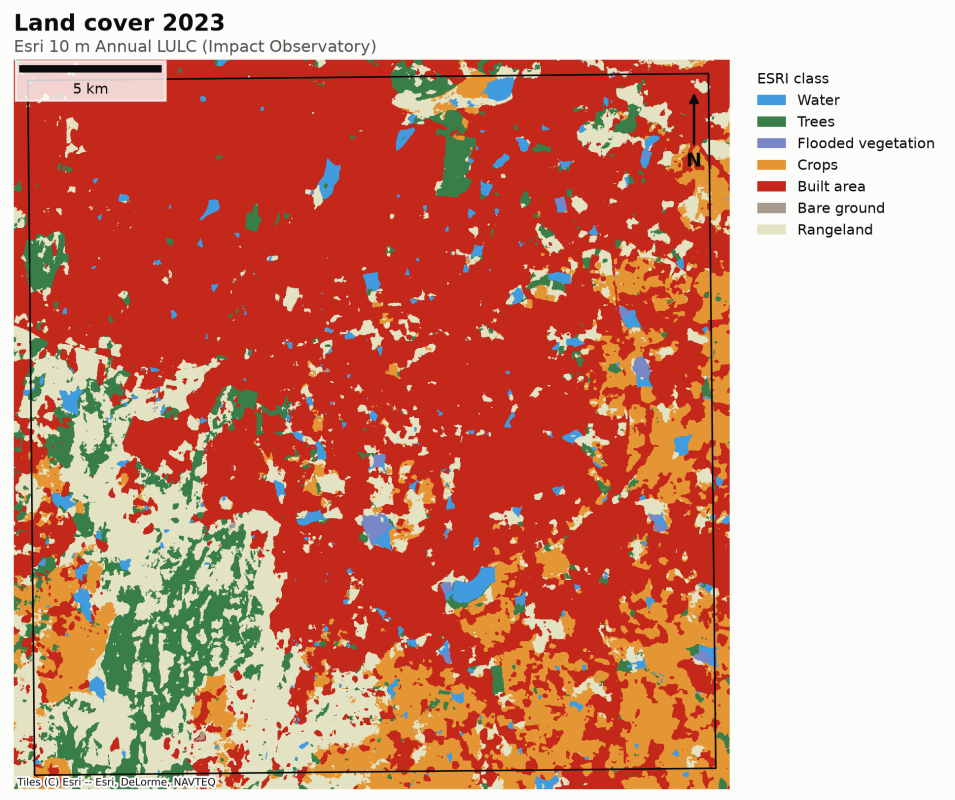


Figure 1: Study area: ESRI land cover in 2023 (Karra et al., 2021), with the area outline. Bannerghatta National Park is the tree and rangeland block in the south-west.

4 Data

All data are free and openly licensed (Table 2). Rasters are read from the Microsoft Planetary Computer, and only the window needed for the study area plus a 3 km buffer is downloaded. The buffer equals the distance cap of the features, so distances from cells near the edge are exact. OpenStreetMap layers come from the Overpass API through OSMnx (Boeing, 2017). Every download is recorded with its checksum and extent in a manifest, so a rerun reproduces or skips it.

Table 2: Datasets.

Dataset	Years	Resolution	Licence	Used for
ESRI 10 m Annual LULC v02 (Karra et al., 2021)	2018–2023	10 m	CC BY 4.0	land-cover features, three-class map, growth labels
ESA WorldCover v200 (Zanaga et al., 2022)	2021	10 m	CC BY 4.0	cross-check of the three-class map
Copernicus DEM GLO-30 (European Space Agency, 2021)	static	30 m	Copernicus DEM licence	elevation, slope
OpenStreetMap (OpenStreetMap contributors, 2026)	2018-01-01, current	vector	ODbL 1.0	roads, water, buildings, protected areas

Land use / land cover. The ESRI (Impact Observatory) map is used for every year and every purpose, so that "built" has the same definition in the baseline and the test years. It has the highest overall accuracy of the global 10 m products (75 %), and built area is one of its most reliable classes (Venter et al., 2022). Its known bias, over-estimating scrub, matters here: much of the Bannerghatta forest appears as rangeland. Inside the park, ESRI classes 47 % of the land as trees and 50 % as rangeland. ESA WorldCover 2021 is used only as an independent cross-check. The two products disagree strongly on built-up land (62 % of the area in ESRI 2023 vs 33 % in WorldCover), because ESRI generalises urban areas to include gardens and street trees. ESRI land cover also flickers between years: only 35–49 % of single-year "new built" pixels are still built later. The growth labels therefore require two years at each end (Section 6).

Terrain. The Copernicus DEM is warped to UTM at 30 m. Slope is computed with Horn's method (Horn, 1981) and then resampled to the 10 m grid. The DEM is a surface model, so heights include buildings and trees; slope is less reliable in dense built-up cells, but it matters most on open candidate land, where this effect is small.

OpenStreetMap. Roads, water and buildings are downloaded twice: as they were on 1 January 2018 (an Overpass `[date:]` query) for the validation run, and as they are now for the final map. The difference is large. All-road length more than doubled between the two snapshots (3889 km to 8730 km), partly new streets in growth areas and partly later mapping. Road features from today's map would therefore leak future growth into a 2018 analysis. OSM building coverage is also very uneven: buildings were mapped in 62–75 % of the built cells in the north but in about 1 % in the south in 2018, and almost all are untagged (`building=yes`). Protected areas (Bannerghatta, tagged as a national park) are a static layer and come from the current map.

Preprocessing. All rasters are aligned to one reference grid in EPSG:32643: 10 m pixels whose edges fall on whole 100 m cells, so every block of 10×10 pixels is one grid cell. Land cover is copied without resampling, because the ESRI tile is already on this lattice. OSM layers are reprojected, clipped to the grid, repaired and split into single-part geometries. Drivable roads are split into major (motorway to secondary) and minor. Buildings smaller than 10 m^2 or larger than 10 ha are dropped. Each preparation step is checked independently against the source data, for example by comparing pixel values at random points, matching road lengths before and after clipping, and overlaying OSM lakes on ESRI water.

5 Methodology

The analysis runs as one pipeline (`python -m src.pipeline`). Every stage reads and writes files with a fixed schema, and every parameter lives in `config/config.yaml`.

5.1 Analysis grid and features

The study area plus a 3 km buffer is divided into 100 m cells aligned with the 10 m raster grid, so every cell holds exactly 10×10 land-cover pixels. The 58,218 cells whose centres lie inside the study area are analysed. For each cell and year the feature table holds:

- **own-cell land cover**: the shares of water, trees, flooded vegetation, crops, built area, bare ground and rangeland among its 100 pixels;
- **context rings**: the same shares in square rings of 250 m and 500 m around the cell, *excluding* the cell itself;

- **distances** from the cell centre to the nearest built-up pixel outside the cell, to water, to a major road and to any road (log-transformed, capped at 3 km, the buffer width, so that edge cells see everything within the cap);
- **road density**: road length within 500 m per km^2;
- **terrain**: mean elevation, mean and maximum slope;
- building count and footprint share (kept for analysis only).

Leakage rule. A model trained to recognise built land could simply read it from the cell's own land cover, whose shares sum to one minus the built share, or from its buildings. In a prototype, own-cell shares alone separated built from non-built cells perfectly. The similarity and the random forests therefore use only the context features: rings, distances, road density and terrain (20 inputs). The code refuses any other column.

5.2 Land classification

K-Means (Pedregosa et al., 2011) groups the cells on seven standardised inputs: the shares of water, trees, crops, built area and rangeland, mean elevation and mean slope. (Snow is absent and flooded vegetation and bare ground are almost always zero; maximum slope correlates at $r = 0.95$ with mean slope.) The number of clusters is chosen from $k = 3$ to 10 with inertia, the silhouette (Rousseeuw, 1987) and the Davies–Bouldin index (Davies and Bouldin, 1979); Gaussian mixtures are compared. Clusters become classes by rules on their centroids: water-dominated → excluded; built-up if built land exceeds 30 % and is the largest land group; forest if trees exceed 50 %, or trees and rangeland together exceed 50 % on slopes above 8° (the Bannerghatta scrub forest, which the ESRI map partly calls rangeland); otherwise usable, with cropland flagged as agricultural.

The **exclusion mask** is applied last, cell by cell: more than half of the cell is water, flooded vegetation or snow in either year of the run, or an OSM lake; mean slope above 15°; more than half no-data; or more than half inside a protected area.

5.3 Building similarity

Reference buildings. Cells with an OpenStreetMap building, plus ESRI built cells where OSM has none: OSM maps 62–75 % of built cells in the north but about 1 % in the south in 2018, so OSM alone would describe only northern surroundings.

Profiles. A location's profile is its cell's 20 context features, standardised with the mean and standard deviation of the reference and candidate cells of the run.

Single query. For building A, every usable cell is ranked by the Euclidean distance between its profile and A's (cosine distance is available for comparison). Each result lists the features where B is closest to A and the ones where it differs most, in standard deviations.

Aggregate similarity. For every candidate cell, the mean distance to its ten nearest reference profiles, never counting the cell itself, is rescaled to 0–1 between the 1st and 99th percentile (higher = more similar).

5.4 Learned models

Two random forests (Breiman, 2001) use the 20 context features of 2018:

- the **change RF** learns from the cells that were non-built in 2018 and 2019 and became built in 2020 and 2021 (355 positives among 15,658 candidates);

- the **state RF** learns built vs non-built cells in 2018 (for comparison).

Both are class-balanced (300 trees). Hyperparameters are tuned by spatial cross-validation (Roberts et al., 2017; Valavi et al., 2019): cells are grouped into 2 km blocks, which are dealt into five folds, so no block is in both training and test data. Tuning uses only the training label. Features enter as percentiles within their year, because OSM road mapping grew a lot between the 2018 and current snapshots; within a year the transform keeps each tree's partition of its training data. Importance is reported both by impurity and by permutation on the held-out folds.

5.5 MCDA / AHP baseline

The classic overlay rescales five criteria to 0–1: distance to built-up land (1 next to it, 0 beyond 2 km), distance to a major road (0 beyond 3 km), slope (0 at 15°), own-cell land cover (rangeland and bare ground 1, crops 0.6, trees 0.2) and a water setback (0 within 100 m, 1 beyond 200 m). Weights come from an AHP pairwise matrix (Saaty, 1980): built-up and road distance 0.30 each, slope and land cover 0.16 each, water 0.09, with a consistency ratio of 0.003. MCDA is not trained, so its use of own-cell land cover is not the leakage problem above.

5.6 Combined score, classes and candidate sites

The final score is a weighted mean of the aggregate similarity and the change RF. Each is first converted to its percentile among the scored cells, because the two have very different distributions. The weight is chosen on the validation run, but on the change RF's training label (growth by 2020/21), never on the later growth used for validation, and then frozen: 0.3 for the similarity and 0.7 for the change RF.

Usable cells are classed by quantiles of the score: S1 at least the 90th percentile, S2 the 70th, S3 the 40th, otherwise N. Forest, built-up and excluded land is always N. Edge-adjacent S1 cells are merged into candidate patches of at least 1 ha, each with its area, mean score, slope, distances and land cover. The top sites are the 20 best patches of at least 5 ha.

5.7 Implementation

Everything is written in Python: rasterio and GeoPandas for the data, scikit-learn for the models, Matplotlib and Folium for the maps; OpenStreetMap data comes through OSMnx (Boeing, 2017). Each stage has unit tests on synthetic data, a phase gate on the real data, and an independent check script that recomputes its output another way (for example pixel values against the source rasters, brute-force distances, or AUC from ranks). The whole pipeline runs in about 12 minutes after the first download, and a fresh clone reproduces the validation run exactly.

Live demo. The results are also published as a static website:

https://iamabhinav-dev.github.io/gis-urban-suitability/

It shows the suitability map with the top 20 sites, an interactive map of every layer, and a "find places like A" tool. The tool runs the single-building query of Section 5.3 in the browser on the exported cell profiles; a check script confirms that it returns the same top 20 locations as the Python query.

6 Validation design

The question is not whether a score reproduces today's land cover, which is easy, but whether it would have pointed to the land that was built on next. Every score is therefore computed twice.

Two runs. The **validation run** uses only data available in 2019 or earlier: ESRI land cover of 2018, OpenStreetMap as of 1 January 2018, reference buildings of 2018, and a three-class map and exclusion mask built from 2018 and 2019. Its scores are tested against the growth that followed. The **final run** applies the same models to the latest data (2023 land cover, OSM as of 1 October 2026) to produce the suitability map. Two layers are treated as static and used in both runs: the DEM, and the boundary of the national park.

Growth labels. ESRI land cover flickers between years: only 35–49 % of the pixels that turn built in one year stay built. Labels therefore require two years at each end. A **candidate** cell has a built share below 10 % in both 2018 and 2019 and is not excluded. It **grew** if its built share is at least 50 % in both 2022 and 2023. Candidates that end up in between, or flicker, are left out (2,881 cells). This gives 12,777 candidate cells, of which 1,125 grew (8.8 %). Each grown cell is also typed by the landscape expansion index (Liu et al., 2010): *adjacent* growth touches built-up land that existed in 2018, *outlying* growth does not.

Leakage rules.

- No model sees a cell's own land cover or buildings (Section 5).
- Nothing dated after 2019 feeds a validation-run score.
- The change RF learns from growth by 2020/21, and most of those cells also count as growth by 2022/23. Its validation scores are therefore **out of fold**: each cell is scored by a model trained without its 2 km block. In-sample scores would reach AUC 0.997.
- The combined score's weight is chosen on the 2020/21 label, not on the test label.

Metrics. One harness scores every model the same way:

- ROC-AUC on the 12,777 candidate cells;
- precision, recall and **lift** (precision divided by the 8.8 % growth rate) in the top 10 % and 30 % of scores;
- the total operating characteristic (Pontius Jr. and Si, 2014), as reported by Ahmadlou et al. (2016);
- AUC separately for adjacent and outlying growth;
- for the combined score, the share of growth that falls in classes S1 and S2.

Random scores and the distance to built-up land are always reported next to the models: in a prototype, distance to built-up alone reached AUC 0.75, so it is the bar to beat. The targets set in advance were a best model at least as good as this baseline and a lift of at least 2 in the top 10 %.

Further checks.

- **Ablation:** each feature group is dropped in turn.
- **Held-out buildings:** reference cells hidden by spatial block must be ranked high by the rest.
- **Self-query:** querying a building must return its own location near the top.
- **Sensitivity:** the analysis is repeated with slope thresholds of 10° and 20°, $k = 5$ to 8, and cells of 50 and 200 m.

7 Results

7.1 Land classification

Choosing the clusters. K-Means and Gaussian mixture models were run for $k = 3$ to 10 on seven standardised inputs: the cell's shares of water, trees, crops, built and rangeland, its mean elevation and its mean slope (Figure 2). K-Means separates the cells far better at every k: silhouette 0.43–0.61 (Rousseeuw, 1987), against 0.26–0.36 for the mixture models, which also have a worse Davies–Bouldin index (Davies and Bouldin, 1979) at every k. The silhouette of K-Means is flat from $k = 5$ to $k = 8$ and drops sharply at $k = 9$. We use $\boldsymbol{k = 7}$ (silhouette 0.606 on the 2023 features, 0.585 on 2018), which is the best or within 0.003 of the best in both years. The seven clusters are the five land-cover groups plus two sub-groups that appear in both years:

- **steep natural vegetation** (mean slope 13.2°, 3 % of the cells): the Bannerghatta hills, mostly ESRI rangeland;
- **the low western valley** (mean elevation 759 m, 3 % of the cells), where cropland and built land are mixed.

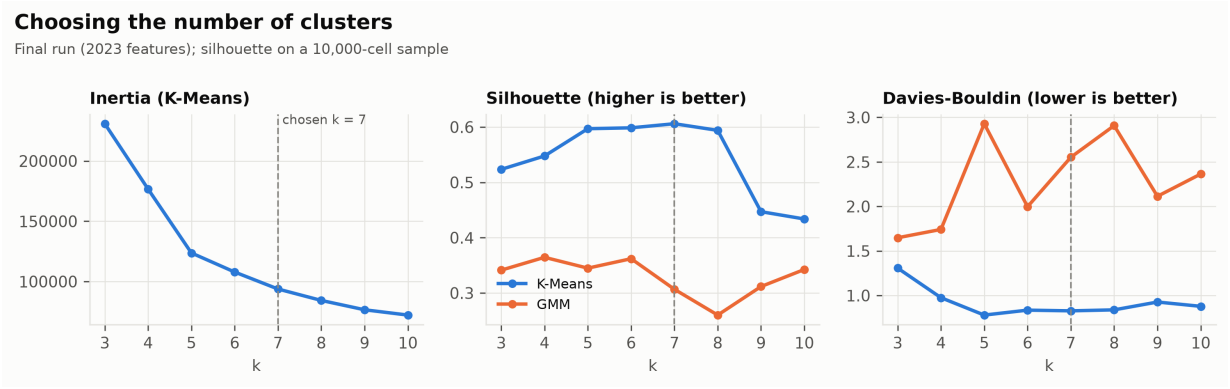


Figure 2: Choosing the number of clusters (2023 features). The dashed line marks $k = 7$.

From clusters to classes. Each cluster is labelled from its centroid (Figure 3): water-dominated clusters are excluded; built-up where built land is both above 30 % and the largest land group; forest where trees exceed 50 %, or where trees and rangeland together exceed 50 % on slopes above 8°; everything else is usable, with cropland-dominated clusters flagged as agricultural. The second forest rule is what keeps the Bannerghatta scrub forest out of the usable class. Finally, the exclusion mask (water, wetland, slopes over 15°, the national park) is applied cell by cell.

The three-class map. Figure 4 shows the result for 2023. Built-up land covers 348 km^2, usable land 138 km^2, forest 29 km^2, and 66 km^2 are excluded. Between the 2018 and 2023 maps, built-up land grows by 42 km^2 and usable land shrinks by 44 km^2 (Figure 5).

Against ESRI's own classes, collapsed to the same three, the map agrees on 96.6 % of the land cells. This is expected, since the ESRI fractions are inputs, and it mainly shows that the cluster rules don't distort them. Against the independent WorldCover map the agreement is 64.0 %. Almost all of the difference comes from the products themselves: WorldCover calls only 33 % of the area built-up, against 62 % in ESRI.

7.2 Building similarity

Single-building queries. Figure 6 shows two of the four example queries from notebook 05, for a school and an industrial building. The most similar usable cells lie at the edges of the built-up

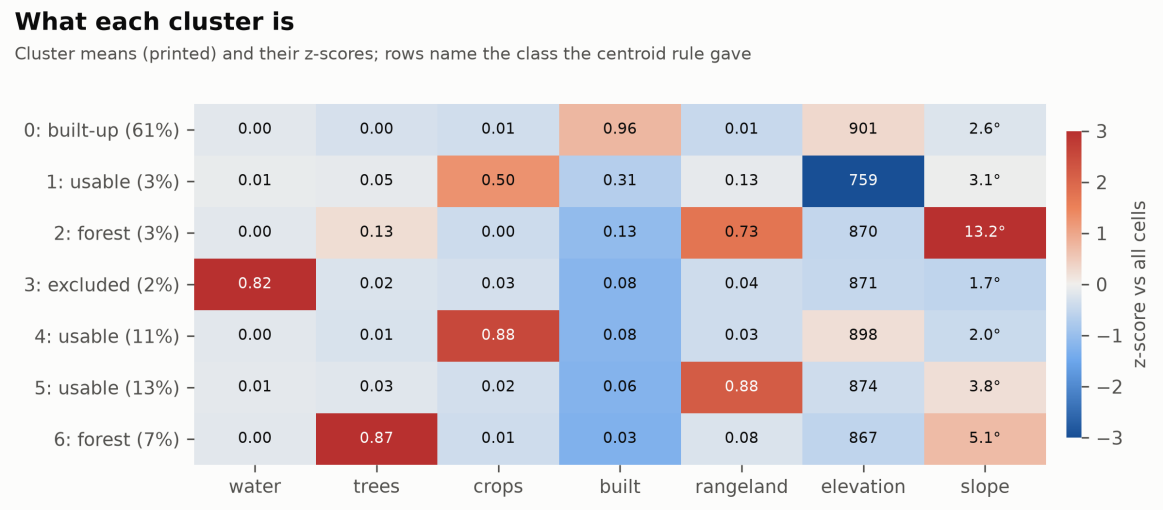


Figure 3: Cluster centroids (2023): mean shares, elevation and slope, coloured by z-score. Each row is named with the class its rule gave and its share of the cells.

area, often several kilometres from A, wherever the mix of built-up land, roads and terrain in the 250 and 500 m rings matches A's. Each result comes with an explanation: for the industrial building's best match, the distances to built-up land, water and a major road differ by less than 0.1 standard deviations, while the built-up share within 250 m differs most (0.6 standard deviations: 80 % instead of 100 %). Land-cover shares that are zero at both cells are listed last, since such matches say nothing. Euclidean and cosine distance share 14–18 of the top 20 cells; cosine ignores how extreme a profile is, so it is kept only as a comparison.

Sanity checks. For 200 random buildings, a cell within one cell of the building is in the top 1 % of all cells even when the building's own cell is left out: surroundings vary smoothly in space, so the profile identifies its location. Reference cells held out by 2 km block are ranked above the other cells with AUC 0.895 by the nearest-neighbour similarity, 0.984 by a random forest and 0.887 by distance to built-up alone. Keeping both ring scales beats the 500 m rings alone (0.895 vs 0.871).

Aggregate similarity. Averaged over the ten nearest reference profiles, the similarity of every candidate cell reaches **AUC 0.773** on the growth by 2022/23 (Table 4), above distance to built-up land, although it never sees a growth label.

7.3 Learned models

The **change RF** reaches **AUC 0.779**, slightly above distance to built-up land (Table 4). Its most important input is the built-up share within 500 m, followed by trees and crops nearby and the distance to built-up land; terrain and water add little (Figure 7). Scored on its own training cells, it would claim AUC 0.997: most of its 2020/21 positives also grew by 2022/23, so in-sample scores would leak the answer, and out-of-fold scoring is essential. The **state RF** separates built from non-built cells almost perfectly (spatial-CV AUC 0.998), but that skill does not transfer to predicting the next growth (0.702).

7.4 Combined score, classes and candidate sites

The weight chosen on the 2020/21 training label is 0.3 for the similarity and 0.7 for the change RF. In the final run, the 13,814 usable cells fall into 1,382 S1, 2,762 S2, 4,144 S3 and 5,526 N cells (Figure 8). Highly suitable land concentrates on the southern and eastern fringe of the built-up area, along the Hosur Road corridor, and in the low western valley.

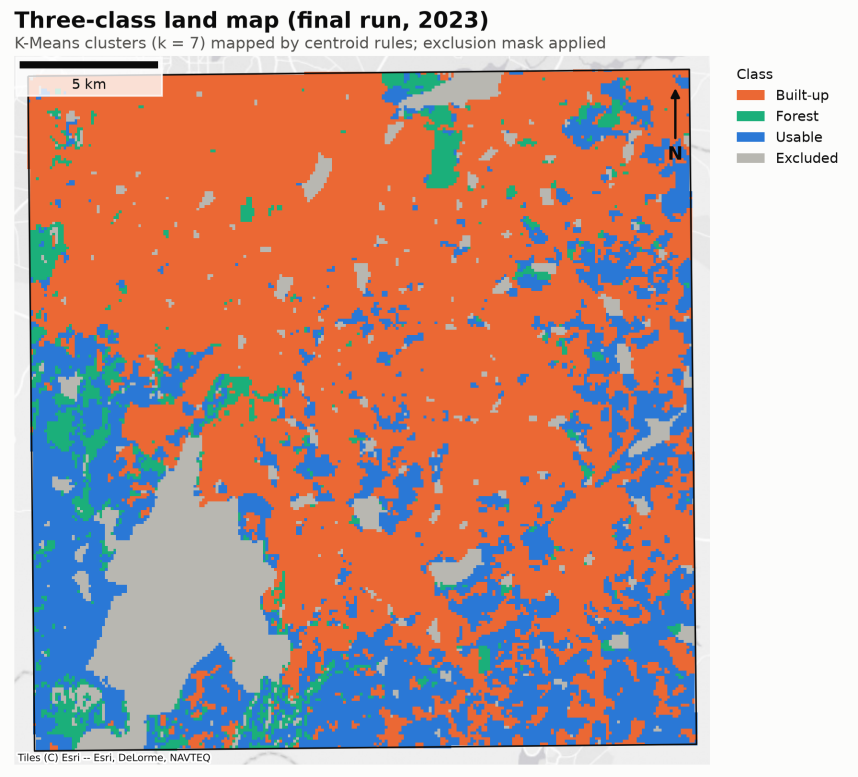


Figure 4: Three-class land map, 2023 (final run).

Merging edge-adjacent S1 cells gives 557 candidate patches covering 1,382 ha; 253 are mostly cropland and flagged as agricultural. Half of the patches are single 1 ha cells, often pockets inside built-up areas, so the top sites are taken from patches of at least 5 ha. The 20 top sites (Table 3, 138 ha) are all gently sloping (1°–4°), directly next to existing built-up land, and covered by rangeland or crops. None of them overlaps an OSM lake (0.06 ha in total) or the national park.

Table 3: The ten best candidate sites (patches of at least 5 ha, ranked by mean score). Distances are means over the patch's cells.

Rank	Area (ha)	Score	Slope (°)	Major road (m)	Built-up (m)	Cover	Location
1	5	0.98	1.9	371	56	rangeland	77.64 E, 12.86 N
2	6	0.98	1.6	1,621	55	rangeland	77.69 E, 12.78 N
3	5	0.97	1.9	226	55	crops	77.73 E, 12.86 N
4	5	0.97	3.9	572	57	rangeland	77.53 E, 12.90 N
5	6	0.96	1.8	358	55	rangeland	77.71 E, 12.76 N
6	6	0.96	1.6	161	55	rangeland	77.70 E, 12.86 N
7	7	0.96	1.5	503	55	rangeland	77.66 E, 12.76 N
8	6	0.96	1.5	1,265	55	crops	77.67 E, 12.79 N
9	6	0.96	1.4	331	55	rangeland	77.71 E, 12.74 N
10	5	0.95	1.3	412	61	rangeland	77.73 E, 12.79 N

7.5 Validation of all scores

Table 4 compares every score on the same 12,777 candidate cells, of which 1,125 were built in both 2022 and 2023. The **combined score** is the best: AUC 0.797, and its top 10 % of cells contain 3.45 times as much growth as a random tenth (ROC and TOC curves in Figures 9a and 9b). Both PRD targets are met (best model at least as good as distance to built-up; lift of at least 2 in the top 10 %). The S1 and S2 classes of the validation run hold 65 % of the growth on 25 % of the

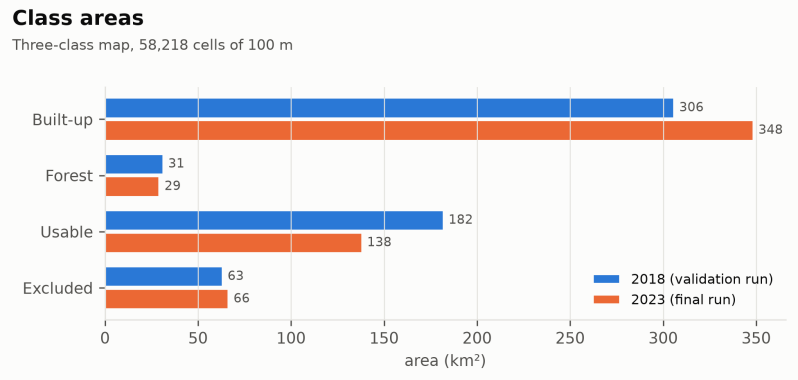


Figure 5: Class areas in the 2018 (validation) and 2023 (final) maps.

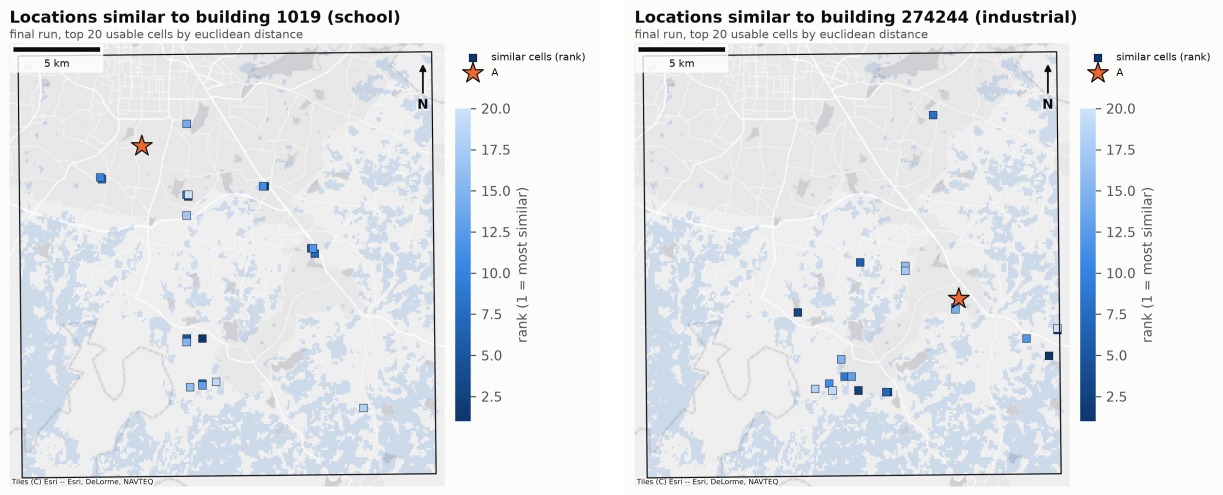


Figure 6: Building similarity queries: the 20 usable cells whose surroundings are most similar to a school (left) and an industrial building (right), A marked with a star.

evaluated cells.

Growth type. Every score finds growth next to existing built-up land well (AUC 0.79–0.82 for the best ones) but growth away from it only at chance level (0.51); distance to built-up and MCDA even fall below chance there (Figure 10a). This matches the finding that diffusional growth is much harder to predict than aggregated growth (Wang et al., 2021). Only 91 of the 1,125 grown cells are outlying, so this hardly moves the overall AUC.

Ablation. Dropping the built-up context (built-up shares in the rings and distance to built-up) costs the most: −0.034 AUC for the change RF and −0.057 for the similarity (Figure 10b). Roads and water add nothing once the built-up context is known.

7.6 MCDA / AHP baseline

The classic method uses five criteria, each rescaled to 0–1: distance to existing built-up land, distance to a major road, slope, land cover (rangeland and bare ground easiest to develop, then crops, then trees) and a setback from water. The AHP pairwise matrix gives built-up and road distance 0.30 each, slope and land cover 0.16 each and water 0.09. Its consistency ratio of 0.003 is far below Saaty's 0.1 limit (Saaty, 1980).

Tested against persistent growth from 2018 to 2022/23, the overlay reaches **AUC 0.636** (Table 4), with a lift of only 1.86 in the top 10 %, below a single criterion, distance to built-up

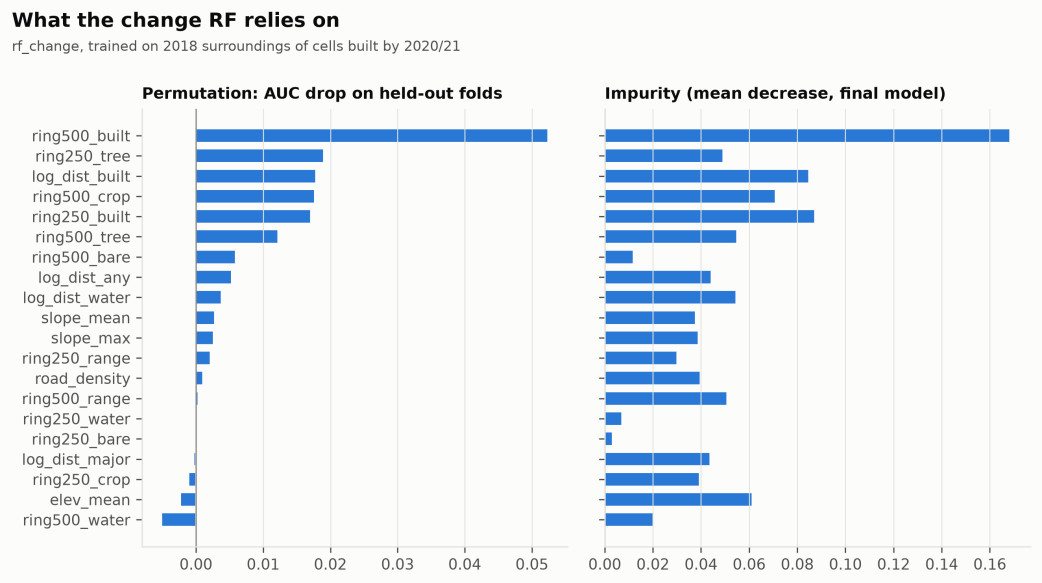


Figure 7: What the change RF relies on: permutation importance (AUC drop on held-out folds) and impurity importance.

Table 4: Validation run: every score computed with 2018 data, tested on persistent growth by 2022/23 (12,777 candidate cells, 1,125 grew). RF scores are out of fold.

Score	AUC	Lift top 10 %	Lift top 30 %	AUC adjacent	AUC outlying
Combined score	**0.797**	**3.45**	**2.40**	**0.822**	0.512
Change RF	0.779	3.23	2.27	0.803	0.512
Building similarity	0.773	2.99	2.22	0.796	0.511
Distance to built-up (baseline)	0.763	3.04	2.15	0.792	0.439
State RF	0.702	3.29	1.92	0.722	0.484
MCDA / AHP	0.636	1.86	1.64	0.655	0.420
MCDA without land cover	0.614	1.73	1.46	0.630	0.430
Random	0.499	1.00	1.02	0.501	0.477

land (0.763). Scored one at a time, only that criterion carries real signal: slope reaches 0.612, and roads, land cover and water stay between 0.52 and 0.55, close to chance. The AHP weights give the weak criteria 70 % of the weight, which dilutes the one strong signal. We deliberately did not tune the weights on the growth label, which is the test set. A fixed, defensible expert overlay is exactly what this baseline stands for, and it illustrates the weakness the literature describes: the weights decide the map, and nothing checks them against outcomes (Malczewski, 2004).

7.7 Sensitivity

Three settings were varied (Figure 11):

- **Slope threshold** (10°, 15°, 20°): the number of slope-excluded cells changes from 1,685 to 158, but every AUC moves by less than 0.01.
- **Number of clusters** ($k = 5$ to 8): class shares change by about one percentage point and agreement with ESRI stays at 96.6–98.8 %. $k = 5$ agrees slightly better with ESRI but loses the steep-vegetation cluster, so forest falls from 5.0 % to 3.9 % of the area.
- **Cell size** (50, 100, 200 m; features and labels rebuilt each time): the change RF scores 0.797, 0.779 and 0.780, distance to built-up 0.769, 0.763 and 0.778, and MCDA 0.631, 0.636 and 0.647. The growth label itself changes with the cell, and at 200 m only 148 cells grew, so differences of a few thousandths are noise.

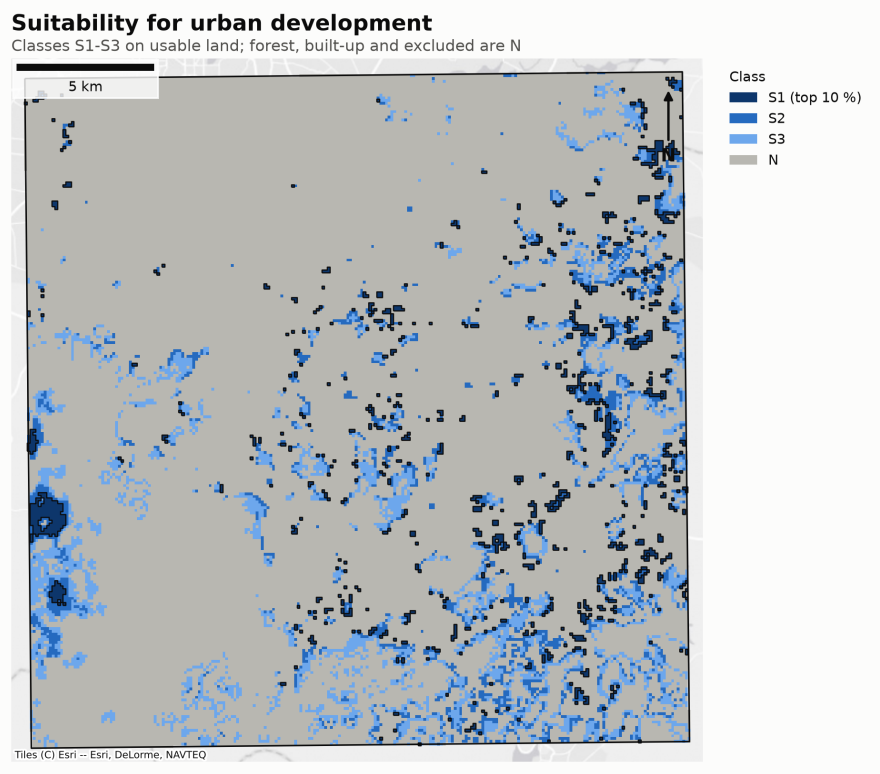


Figure 8: Suitability classes in the final run (2023 data) with the outlines of the candidate patches. Forest, built-up and excluded land is always N.

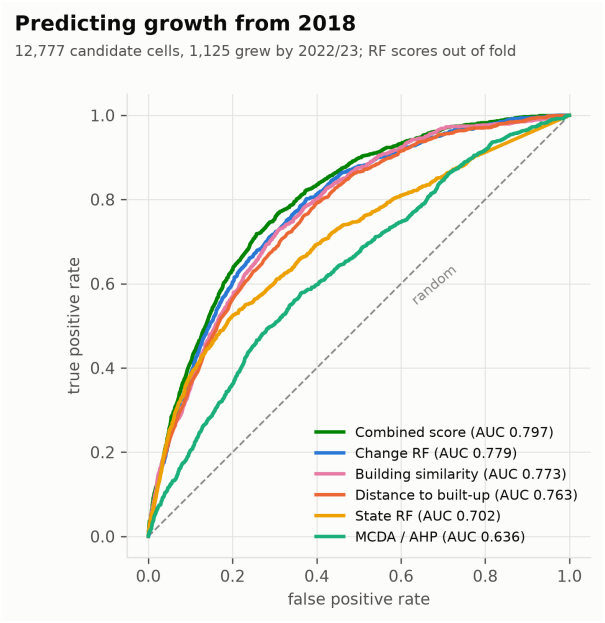
What holds in every setting is the ranking: **change RF > distance to built-up > MCDA**.

8 Discussion and limitations

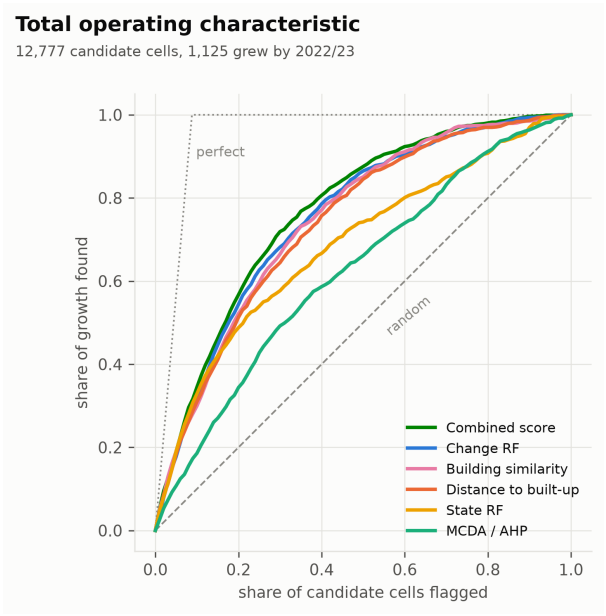
Learned scores beat the expert overlay, but not by much against the simplest rule. The combined score improves on distance to built-up land by 0.034 AUC and on the classic AHP overlay by 0.161. The first gap is modest because growth in Bengaluru South is overwhelmingly edge expansion: 92 % of the grown cells touch land that was already built in 2018, and a single distance already ranks such cells well. The value of the learned scores lies in the ranking *among* cells next to built-up land, where they add the composition of the neighbourhood (how much built-up, cropland and scrub lies within 250 and 500 m) and terrain. Our AUC is close to the 0.82 reported for a random forest tested on later change in Rasht (Ahmadlou et al., 2016), and far below the near-perfect values of studies that validate on a random split of a single date (Li et al., 2024): the difference comes from how the test is set up, not from a better model.

Why the expert overlay does worse than one of its own criteria. Only one of the five MCDA criteria carries real signal on its own. The pairwise comparisons are internally consistent (CR 0.003), but consistency says nothing about relevance: they put 70 % of the weight on criteria that are close to random for this question. This is the subjectivity Malczewski (2004) and Mosadeghi et al. (2015) warn about, made measurable by a temporal test.

Similarity without labels works. The building similarity never sees a growth label, yet it predicts growth almost as well as the random forest trained on recent change (0.773 vs 0.779), and their combination is better than either. It also explains itself: each match comes with the features where A and B agree and where they differ, which a planner can check.

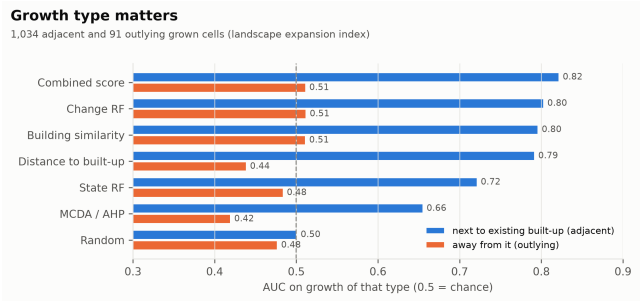


(a) ROC curves.

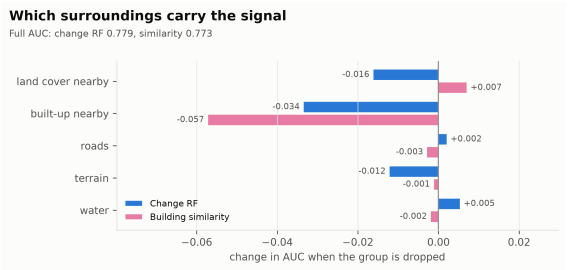


(b) Total operating characteristic.

Figure 9: Predicting growth from 2018.



(a) AUC by growth type.



(b) AUC change when a feature group is dropped.

Figure 10: Where the models succeed and what they rely on.

Outlying growth is not predicted. No score does better than chance for growth away from existing built-up land, as Wang et al. (2021) found for diffusional growth. Such growth follows decisions the data do not record: land deals, new industrial estates, approvals.

Limitations.

- **Land-cover data.** ESRI and WorldCover disagree on built-up land (62 % vs 33 % of the area): ESRI generalises urban areas, and it calls part of the Bannerghatta forest rangeland (Venter et al., 2022). Growth labels inherit its errors; the two-year rule removes flicker but not systematic bias.
- **OpenStreetMap.** Road mapping more than doubled between the snapshots, mostly through mapping effort, so each run uses its own snapshot. OSM buildings were sparse in the south in 2018 (Herfort et al., 2023) and almost none are typed, so similarity per building type was not possible.
- **Resolution.** The 100 m cells mix land uses; half of the top-class patches are single cells inside built-up areas. Repeating the analysis at 50 and 200 m changes the AUCs by at most 0.02, but the growth label itself depends on the cell size.

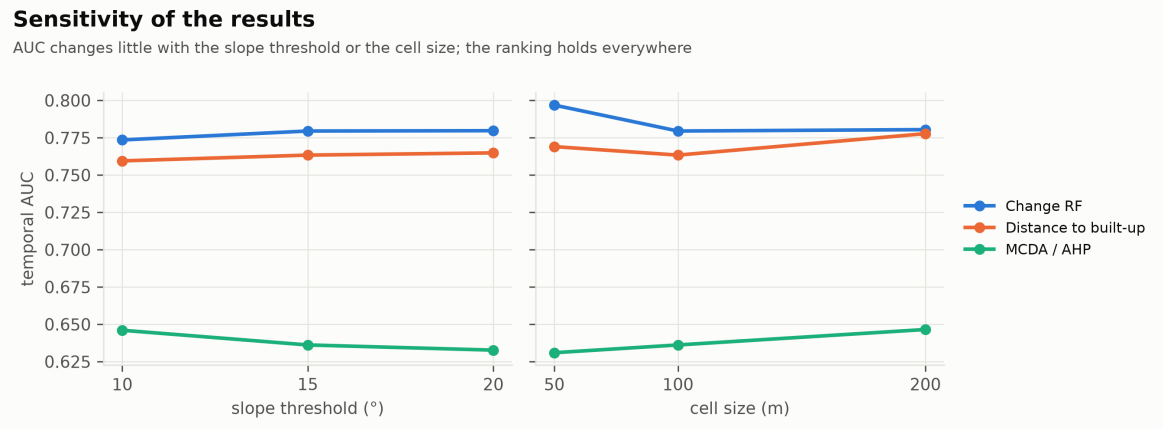


Figure 11: Temporal AUC by slope threshold and by cell size.

- **Past growth is not good planning.** Every score learns where building *has* happened. Water, steep slopes and the national park are excluded, but zoning, flood risk, infrastructure capacity and ownership are not in the data. The suitability map shows where growth is *likely*, as an input to planning, not a recommendation.

9 Conclusion and future work

This project built a land suitability analysis for urban development in Bengaluru South entirely in Python, from open land-cover, terrain and OpenStreetMap data, and tested every method against the growth that actually happened. It answers the three questions it set out with:

1. **Land classes.** Clustering 100 m cells on land-cover shares and terrain gives seven interpretable clusters, mapped to built-up (348 km^2 in 2023), forest (29 km^2, including the steep scrub forest of Bannerghatta) and usable land (138 km^2). The map agrees with the ESRI classes on 96.6 % of land cells.

2. **Building similarity.** For any building, the usable locations whose surroundings look most like its own can be found and explained feature by feature. Averaged over reference buildings, this similarity predicts growth from 2018 to 2022/23 with AUC 0.773, without ever seeing a growth label.

3. **Suitability.** Combining the similarity with a random forest trained on recent change gives the best score (AUC 0.797; the top 10 % of cells hold 3.45 times their share of growth), clearly ahead of a classic AHP overlay (0.636) and ahead of the distance to built-up land (0.763). The final map grades usable land into classes S1–S3 and lists the 20 best candidate sites.

The main lesson is about validation: strict rules against leakage (no own-cell land cover, data frozen at the baseline date, out-of-fold scores) turn near-perfect accuracies into honest ones, and simple baselines show how much of the signal is just "next to existing built-up land".

Future work.

- Add drivers of growth away from the built-up edge: planned layouts, industrial estates, land prices, distance to the city centre and to employment.

- Re-run the analysis with newer years as they become available, moving the baseline forward to test whether the models generalise in time.

- Use building footprints from other sources (e.g. Open Buildings) to type buildings, so that similarity can be computed per building type.
- Test finer grids with a growth label defined independently of the cell size.

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